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Does compensation improve tolerance of wild boars?

Understanding the compensation policy impact on local resident's tolerance towards wild boars in Giant Panda National Park, China

Statistical Consulting
LMU Munich
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Outline

1. Background
2. Data and EDA
3. Feature Engineering
4. Main Modelling
5. Robustness
6. Multi-Group Analysis
7. Conclusion

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Motivation

- **HWC creates both ecological and social challenges**, including crop damage, livestock losses and threats to human safety (Madden, 2004; Zimmermann et al., 2020).
- **Local tolerance toward wildlife is central to long-term coexistence** and has become an important research focus in HWC management (Expósito-Granados et al., 2019; Kansky & Kidd, 2024).
- **Tolerance depends on more than economic loss** — intangible costs such as fear, perceived risk and attitudes toward wildlife also matter (Kansky & Knight, 2014).
- **Compensation policies aim to offset losses and improve tolerance**, but whether they actually achieve this remains an open question (Hemminger et al., 2025; Kichloo et al., 2023).

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Three Research Questions

- **RQ1:** Does **compensation** for wild boar damage have a **direct effect** on **tolerance** toward wild boar, or is its influence mediated by other social and psychological factors?
- **RQ2:** What roles do **tangible costs, intangible costs,** and **intangible benefits** play in shaping tolerance, and how do these factors **interact** with **compensation** experience (CE)?
- **RQ3:** How **robust** are these relationships under **different model specifications** and **across groups** with and without compensation experience?

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The Questionnaire

Cross-Sectional Household Survey: Each respondent was interviewed once.
Data were collected in Baoxing (Aug 2024) and Qingchuan (Mar–Apr 2025); the dataset is not longitudinal.

1. **Sociodemographics:** Age, gender, education, annual income, household assets, cropland area
2. **Conflict Experience:** Frequency, duration, severity, species diversity, years exposed
3. **Economic & Intangible Costs:** Estimated damaged crop area and economic loss (**tangible**) & attitude toward wild boar occurrence and three-word description (**intangible**)
4. **Compensation & Mitigation:** Application for compensation, compensation amount satisfaction, mitigation strategies used, willingness to pay for insurance
5. **Tolerance:** Four Likert items (1-5): support for non-lethal management, agreement that government should control boars, perceived personal safety threat, acceptance of some damage.

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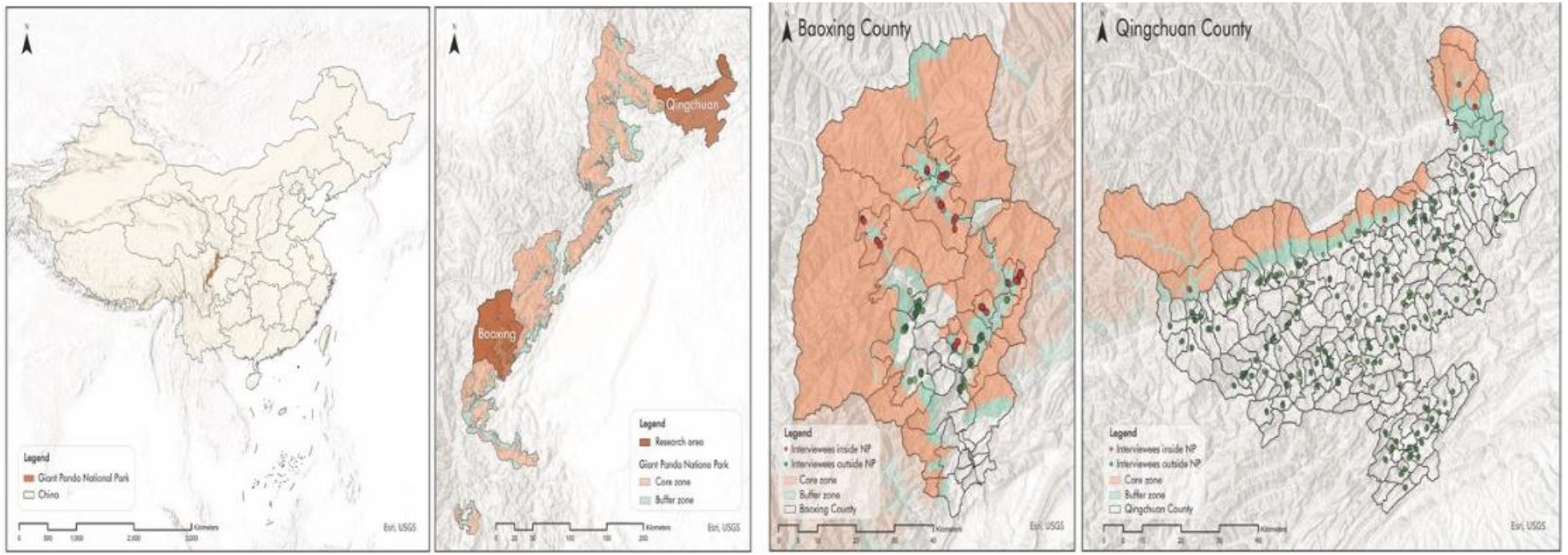
Dataset

- 625 retained respondents with 55 features from two Chinese counties:
  - **Baoxing:** 160 respondents
  - **Qingchuan:** 465 respondents

- Fieldwork documentation places the counties in different survey periods, so place and timing cannot be separated.

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Where are Baoxing and Qingchuan?



- Cross Sectional Study: each interviewee got interviewed only **once**
- In-person, face-to-face surveys, interview language: Mandarin Chinese
- Time period:
  - Baoxing: 2024.08.15 - 2024.08.31 (n=195)
  - Qingchuan: 2025.03.10 - 2025.04.10 (n=479)

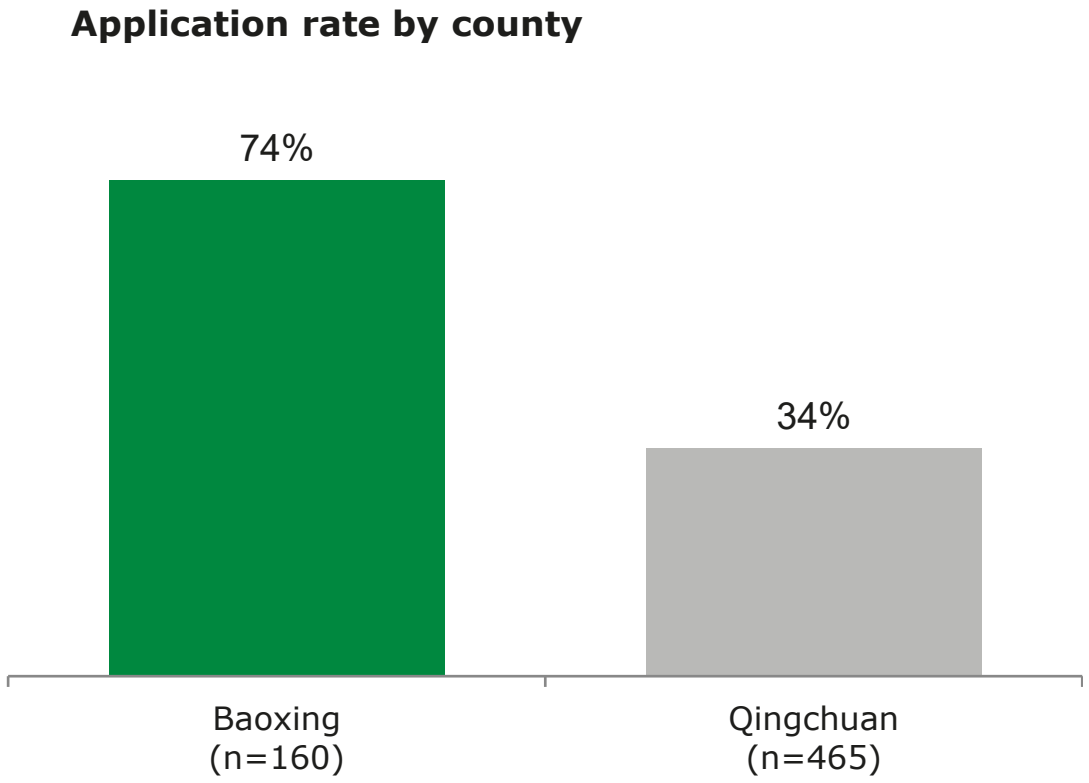
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Compensation participation is confounded with location (county)

- In Baoxing: 119 of 160 respondents (74%) reported applying for compensation.
- In Qingchuan: only 158 of 465 (34%) - a chi-square test confirms this gap is not chance ($\chi^2=77.1$, df=1, $p<.001$).
- The two counties were also surveyed seven months apart: Baoxing in August 2024, Qingchuan in March–April 2025.
- Geography and timing move together with compensation status - any comparison across compensation groups must control for county/survey wave.



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Compensation Recipients differ in Assets, Not in Who They Are

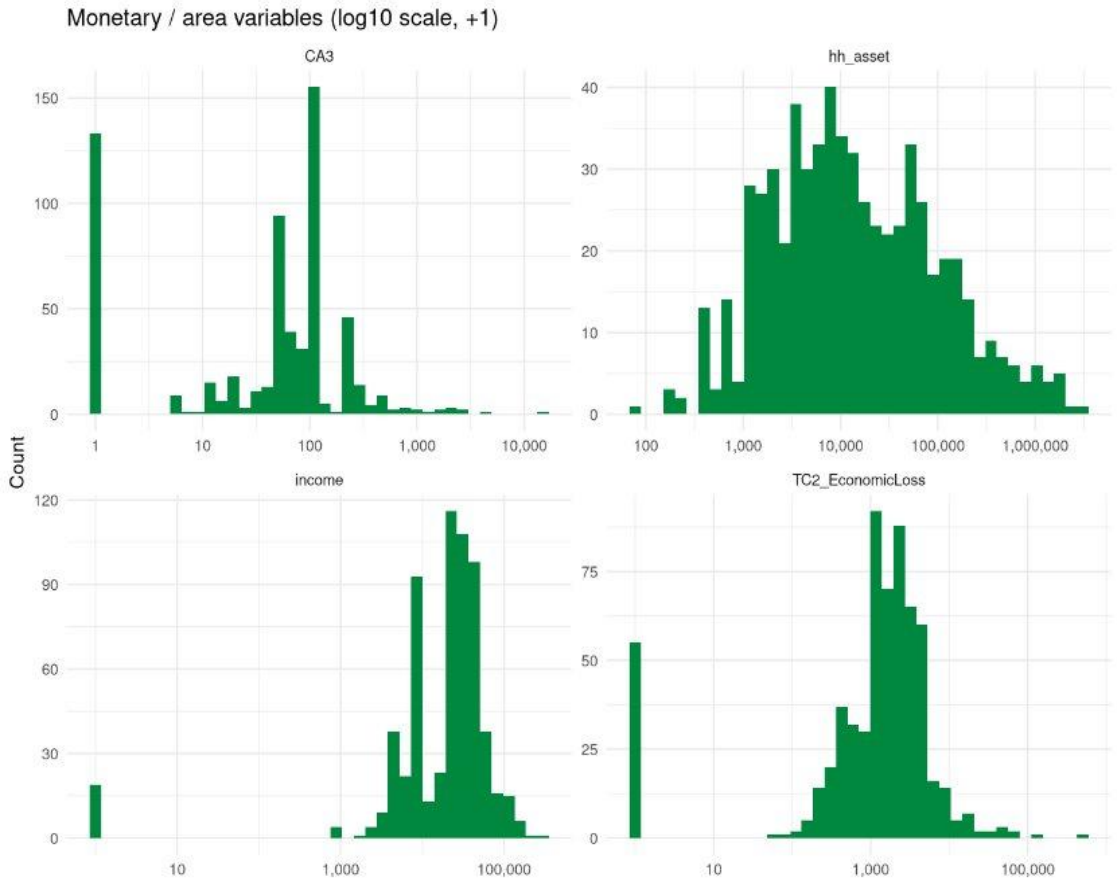
Tests: Wilcoxon rank-sum (Mann-Whitney U) for all variables above except Gender, which uses a chi-square test.

Differs significantly ($p < .05$)		No significant difference	
Total cropland	**$p < .001$**	Age	$p = .401$
Total livestock	**$p < .001$**	Gender	$p = .318$
Household assets	**$p < .001$**	Education	$p = .383$
Agricultural income share	**$p = .017$**	Annual income	$p = .599$

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Skewness, missing values and outliers

- Money/area variables (insurance willingness, assets, income, damage loss) are heavily right-skewed - assets alone range 77 to 3,087,729 RMB.
- All four are log(1+x)-transformed before entering any model, so a handful of large landholders don't dominate the fit.
- Missingness CE1/CE2 are skip-logic (348/625 never applied)
- Out-of-range CA1/CA2 values (9+4 cases) are kept as genuine, not errors
- 14 of 625 respondents (2.2%) are Mahalanobis multivariate outliers (across 15 core Wildlife Tolerance Model (WTM)) indicators (conflict experience, costs, benefits, tolerance items), threshold p<.001) - kept for the main modelling.



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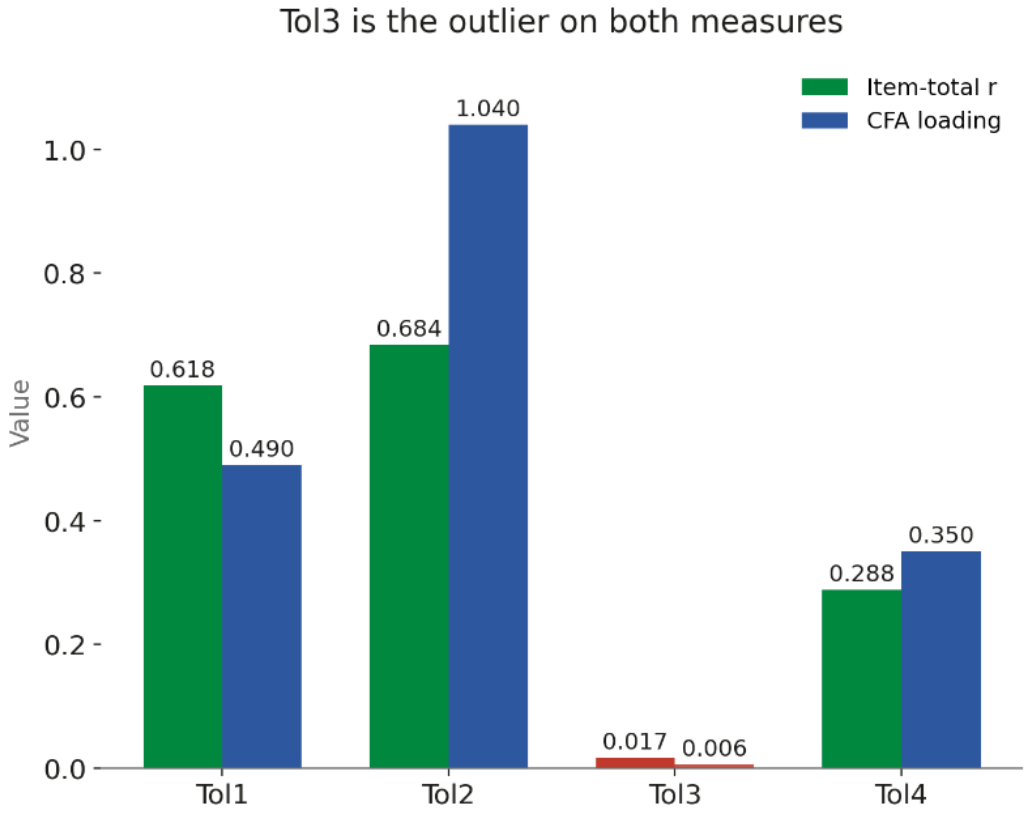
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Tolerance: EFA and CFA both agree 3 indicators

EFA: exploratory factor analysis, R package: psych, GPArotation, nFactors

CFA: confirmatory factor analysis, R packages: lavaan

- The original four-item tolerance scale has low internal consistency: α = 0.34.
- Tol3 is almost unrelated to the other items: item-total r = 0.017, EFA loading=0.006, confirmed by CFA (loading=0.006, p=.90, not different from zero).
  - Tol1) I support non-lethal management strategies for wild boars (e.g., relocation, fencing);
  - Tol2) Government should take stricter measures to control the wild boars population;
  - Tol3) Wild boars cause a serious threat to my personal safety;
  - Tol4) I can accept some damage caused by wild boars.
- Decision: tolerance = Tol1+Tol2+Tol4 (3-item) is the primary construct



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Closer look into intangible cost (IC): formative versus reflective

- A reflective intangible cost specification fails the discriminant-validity check: HTMT (heterotrait-monotrait ratio for correlations) with tolerance = 1.17.

- Intangible cost is therefore retained as a formative composite.

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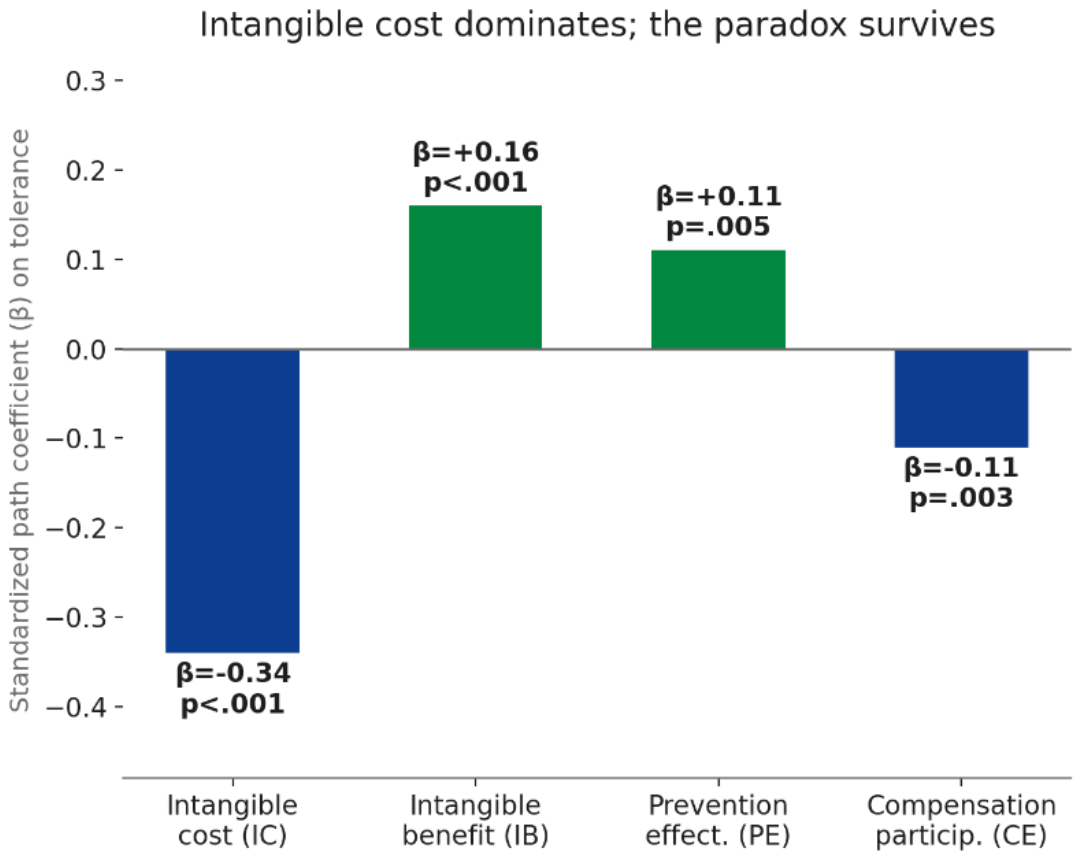
The Model: A full-sample PLS-SEM with 5,000 bootstrap resamples

Reminder RQ1: Does **compensation** for wild boar damage have a **direct effect** on **tolerance** toward wild boar, or is its influence mediated by other social and psychological factors?

- Full sample n=625: tolerance is reflective (Tol1, Tol2, Tol4); costs, conflict and policy are formative — PLS-SEM handles both in one model, using R package seminr
  - Tol1: I support non-lethal management strategies for wild boars (e.g., relocation, fencing);
  - Tol2: Government should take stricter measures to control the wild boars population;
  - Tol4: I can accept some damage caused by wild boars.
- Also Included survey site and national-park residence as controls (compensation differs sharply by survey site(county)); skewed monetary variables enter as log(1+x).
- All paths estimated together with 5,000 bootstrap resamples for uncertainty — results are associations in a cross-sectional survey, not causal estimates.

RQ1&2 Findings

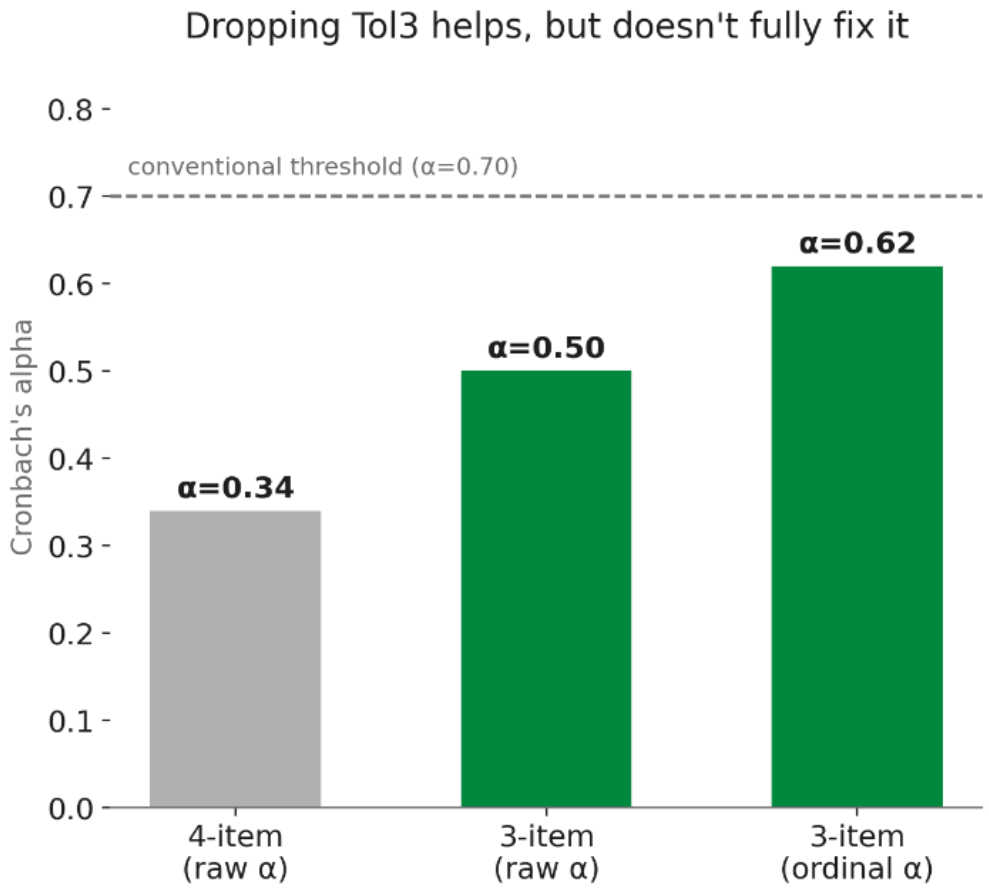
- Intangible cost (IC) → tolerance: β = −0.34, p < .001 - the strongest negative association.
- Intangible benefit (IB) → tolerance: β = +0.16, p < .001
- Prevention effectiveness (PE) → tolerance: β = +0.11, p = .005.
- Compensation participation (CE) → tolerance: β=−0.109, p=.003, even after site adjustment.
- R^2=0.234: meaningful, but much remains unexplained; an association, not proof of causation.



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Some more findings: evidence supports caution

- Dropping Tol3 (“the measurement fix”) raises reliability, but still falls short of the conventional α ≥ 0.70 bar
- PS overlaps with conflict experience & site: HTMT=1.01 and 1.22 - both exceed the 1.0 ceiling => no strong substantive claim on that path alone.
- Survey site construct predicts 5 mediators (p<.05) but not tolerance directly (β=−0.08, p=.34) once the mediators are in the model.



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RQ1: Direct effect or mediation?

- Direct compensation-participation (CE) → tolerance effect: −0.109, p = .003.
- Total indirect effect through five measured mediators:
  - Intangible cost (IC)
  - Tangible cost (TC)
  - Intangible benefit (IB)
  - Policy support (PS)
  - Compensation anticipation (CA)
  - at −0.024: no specific indirect path is significant.
- VAF = 18%: the measured mediation is small

--> The association may reflect unmeasured grievance, prior conflict or another shared cause.

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RQ2: Do costs and benefits work differently after compensation? No.

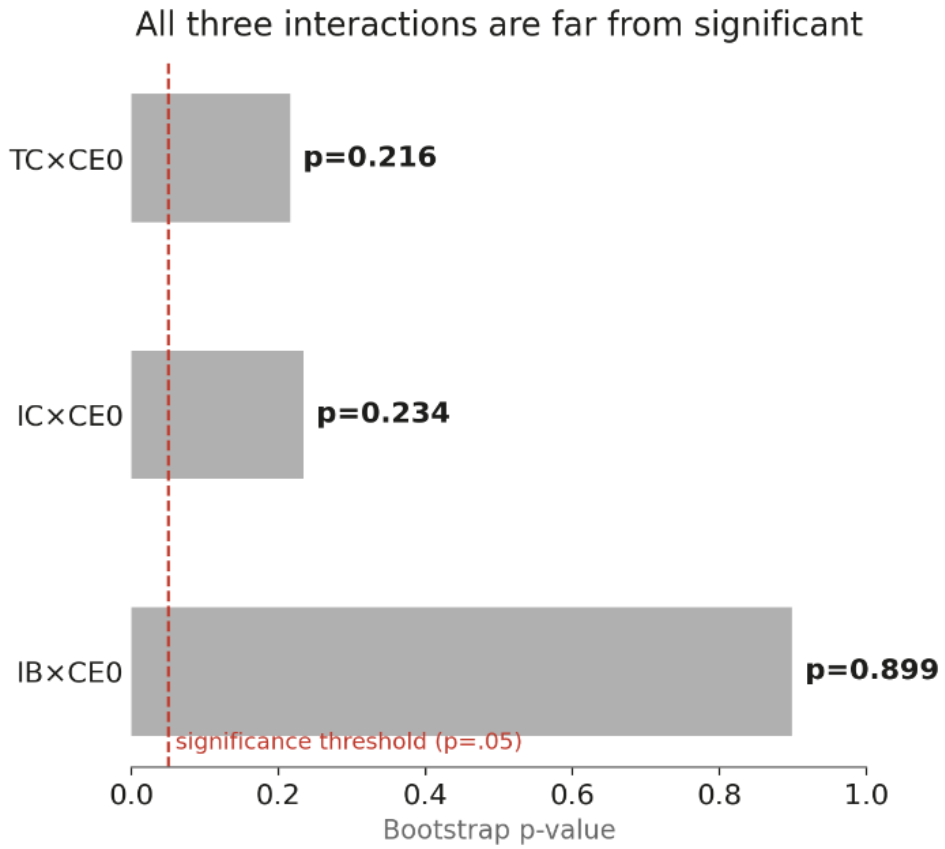
We test this directly with compensation × tangible-cost, intangible-cost and intangible-benefit interaction terms

=> stronger than comparing separate group coefficients, because it asks whether the slopes differ statistically.

Result: all three interactions are non-significant:

- TC×CE0 p=.216
- IC×CE0 p=.234
- IB×CE0 p=.899.

Take away: The cost/benefit → tolerance relationships are the same strength regardless of compensation status.



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In a nutshell: Main-analysis findings

• RQ1: **Compensation Participation** (CE) is negatively associated with **Tolerance**, with **no** established measured **mediator**.

• RQ2: **Intangible Cost** (IC) is the **strongest negative** predictor and **Intangible Benefit** (IB) is **positive**; **Compensation** (CE0) does not alter these slopes.

Outline

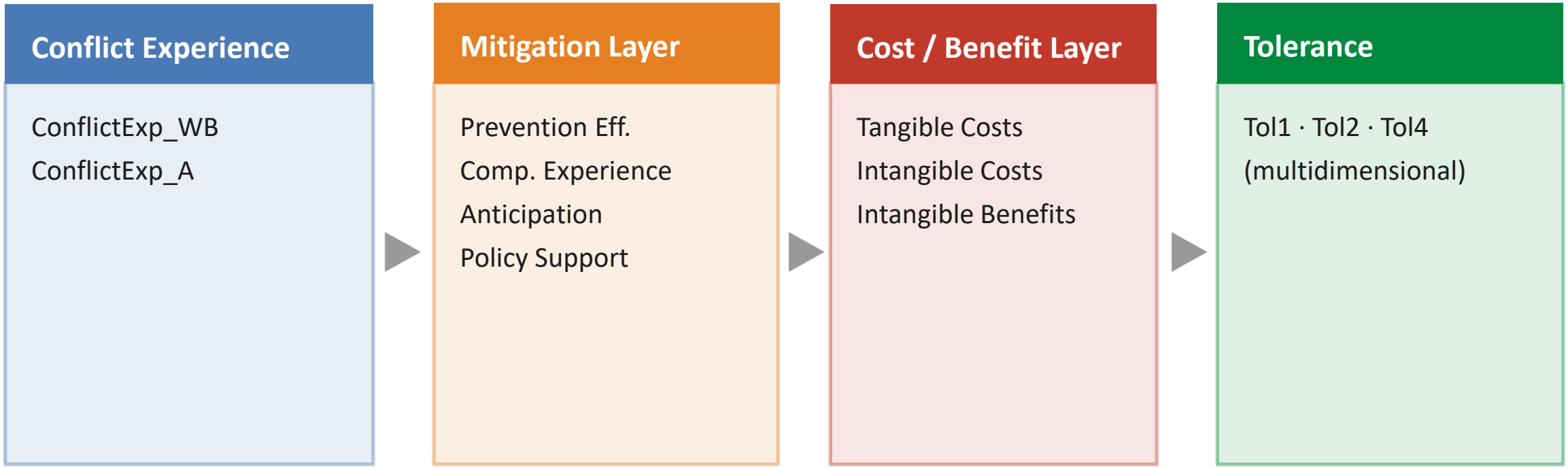
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Why PLS-SEM?

- PLS-SEM handles **formative + reflective** constructs simultaneously
- No multivariate normality assumption required
- Reliable at n = 625 with complex construct structures
- Tolerance AVE = 0.34: **justified** → four attitudinal facets, not one trait

The Wildlife Tolerance Model (WTM)



⚠ *Outer model only: inner factors (values, norms) not captured by the questionnaire.*

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Why Robustness Checks?

"The aim is not to search for significant results, but to examine how sensitive the main findings are to different reasonable analytical choices." - StaBLab Notes

- Try many specs, report only the significant one
- Each individual choice justified
- Result: significant finding that disappears under different choices
- Key question: Does **Compensation → Tolerance stay negative** no matter what?

- **6 theoretically motivated specs**: one analytical choice changed each time
- **8,192 exhaustive models**: every possible combination of 13 adjustments
- Item quality: outer loadings ≥ 0.70 · VIF < 3
- SRMR computed manually across all specs — fit must not degrade

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Six Model Specifications

Model	What changes	Rationale
M0.1 (Base)	All predictors included	Reference model
M_noSocio	Socio-demographics removed	Do demographics inflate the effect?
M_noTangible	Tangible Costs removed	Does tangible cost confound?
M_noPS	Policy Support removed (HTMT = 1.112)	Address discriminant validity concern
M_IC1only	Intangible Costs = IC1 only	Most conservative IC specification
M_TolFull	Tolerance = all 4 items incl. Tol3	Alternative outcome operationalisation

Table 1: Six PLS-SEM Robustness Specifications

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Vibration Of Effects

8,192 *models estimated* *2^{13} combinations*	**100** *Negative 6 across all models*	**99.4%** *Reach $p < .05$*	**0** *Janus effects sign never flip*

- Patel et al. found the **Janus effect** (sign flips) in **31%** of their 417 variables.
- We find it in **zero**.
- Median β = −0.09 across all 8,192 models → consistent with PLS-SEM range (−0.128 to −0.143).

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Multi-Group Analysis

Groups:

- **With compensation** (CE0 = 1): n = 277
- **Without compensation** (CE0 = 0): n = 348
- Split is the central grouping variable of the entire study
- Groups are **unequal** in size

Method: permutation-based MGA

- 1,000 permutations: no normality assumption
- Appropriate for unequal group sizes
- Tests whether path coefficients are **statistically different** across groups
- Goes beyond Duc's visual sub-group comparison

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HTMT Per Group

Before interpreting any path differences, HTMT must be checked **within each subgroup separately**.

If HTMT > 1.00 within a subgroup: Group's path coefficients are unreliable and MGA comparisons would be meaningless.

1. Check HTMT per group
2. Threshold: Violation (> 1.00) within a subgroup invalidates that group's path coefficients entirely
3. Result: Discriminant validity confirmed in both groups

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MGA 1 (full model)

Core finding: structural model is **largely invariant** across compensation groups. Intangible Costs → Tolerance significant in **both groups** ($p < .001$).

Predictor	$\hat{\beta}$ With	$\hat{\beta}$ Without	$\Delta\hat{\beta}$	p	Sig.
Intangible Costs	−0.228	−0.454	0.226	0.131	—
Intangible Benefits	+0.121	+0.220	−0.099	0.289	—
Prevention Eff.	+0.139	+0.086	0.053	0.448	—
ConflictExp_A	−0.186	−0.124	−0.062	0.114	—
Policy Support	−0.082	−0.094	0.012	0.765	—
Tangible Costs	+0.034	+0.016	0.018	0.829	—
HouseholdAsset	+0.091	−0.157	0.248	0.003	* (control)

Table 4: MGA 1 (Full Model): Permutation Test Results ($n_{with} = 277$, $n_{without} = 348$)

R^2: With compensation = 0.299 · Without compensation = 0.423. Permutation test with 1,000 permutations.

HouseholdAsset is a control variable, not a core structural path.

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MGA 2 (No Socio)

Confounder check passed: significant findings in MGA 1 and MGA 2 do not overlap. ConflictExp_A emerges when demographics are removed

Predictor	$\hat{\beta}$ With	$\hat{\beta}$ Without	$\Delta\hat{\beta}$	p	Sig.
Intangible Costs	−0.214	−0.435	0.221	0.112	—
Intangible Benefits	+0.123	+0.273	−0.150	0.167	—
Prevention Eff.	+0.141	+0.082	0.059	0.448	—
ConflictExp_A	−0.187	+0.038	−0.225	0.039	*
Policy Support	−0.078	−0.044	−0.034	0.765	—
ConflictExp_WB	+0.023	−0.126	0.149	0.230	—
Tangible Costs	+0.079	+0.062	0.017	0.829	—

Table 5: MGA 2 (No Socio Controls): Permutation Test Results

Interpretation: compensation experience may amplify the conflict–tolerance link. For compensated residents more conflict → less tolerance (β=−0.187). For uncompensated residents the link is near-zero (β=+0.038).

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F²

Permutation p-value: is the difference statistically significant?

f²: how functionally important is this predictor within each group?

Predictor	f^2 With	Size	f^2 Without	Size
Intangible Costs	0.142	Small	0.237	Medium
Intangible Benefits	0.065	Small	0.148	Small
Prevention Eff.	0.067	Small	0.022	Small
ConflictExp_A	0.041	Small	0.008	Negligible
Policy Support	0.018	Negligible	0.026	Small
Tangible Costs	0.004	Negligible	0.001	Negligible

Table 6: Cohen's f^2 per Group: MGA 1

Cohen (1988) f² thresholds: Negligible < 0.02 · Small ≥ 0.02 · Medium ≥ 0.15 · Large ≥ 0.35

MGA 2 (no socio): Intangible Costs f² = 0.258 (without) vs 0.103 (with) — pattern holds across both MGA versions.

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Structural invariance confirmed: With two findings that matter for RQ3

- **Structural invariance holds:** core paths do not significantly differ between compensated and uncompensated residents across either MGA version; the model is stable
- **RQ3:** Intangible Costs dominates among uncompensated residents
- **ConflictExp_A differs structurally:** $\beta = -0.187$ for compensated residents vs $\beta = +0.038$ for uncompensated ($\Delta\beta = -0.225$, $p = 0.039$ in MGA 2)
- Confounder check passed
- **Lower R^2 in compensated group:** 0.299 vs 0.423 in MGA 1; the model explains uncompensated residents more completely; compensated residents have additional unmeasured

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What limits our conclusions?

- The comparison is cross-sectional; we lack pre-application tolerance and clearly established payment timelines.

- County and survey period coincide, and sampling and earlier exclusions are incompletely documented.

- Short, overlapping measurement blocks and missing word responses limit claims about latent constructs.

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The next study

- Clarify application/payment routing, sampling exclusions, fieldwork dates, area units and percentage anomalies with Mengxi.

- Revise the report transparently and rerun only analyses affected by those clarifications.

- A future panel should measure attitudes before application and after payment, with better indicators and comparison design.

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R-Packages

- Measurement: psych + GPArotation + nFactors for EFA; lavaan + semTools for CFA.
- Structure: factoextra for the PCA index; seminr runs the whole PLS-SEM model — estimation, mediation, moderation, and multigroup analysis together.
- Everything else: dplyr/tidyr/tibble for data wrangling, ggplot2 for figures, all run under R via the targets pipeline.

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Discussion



How is Tolerance (Tol) defined/measure?

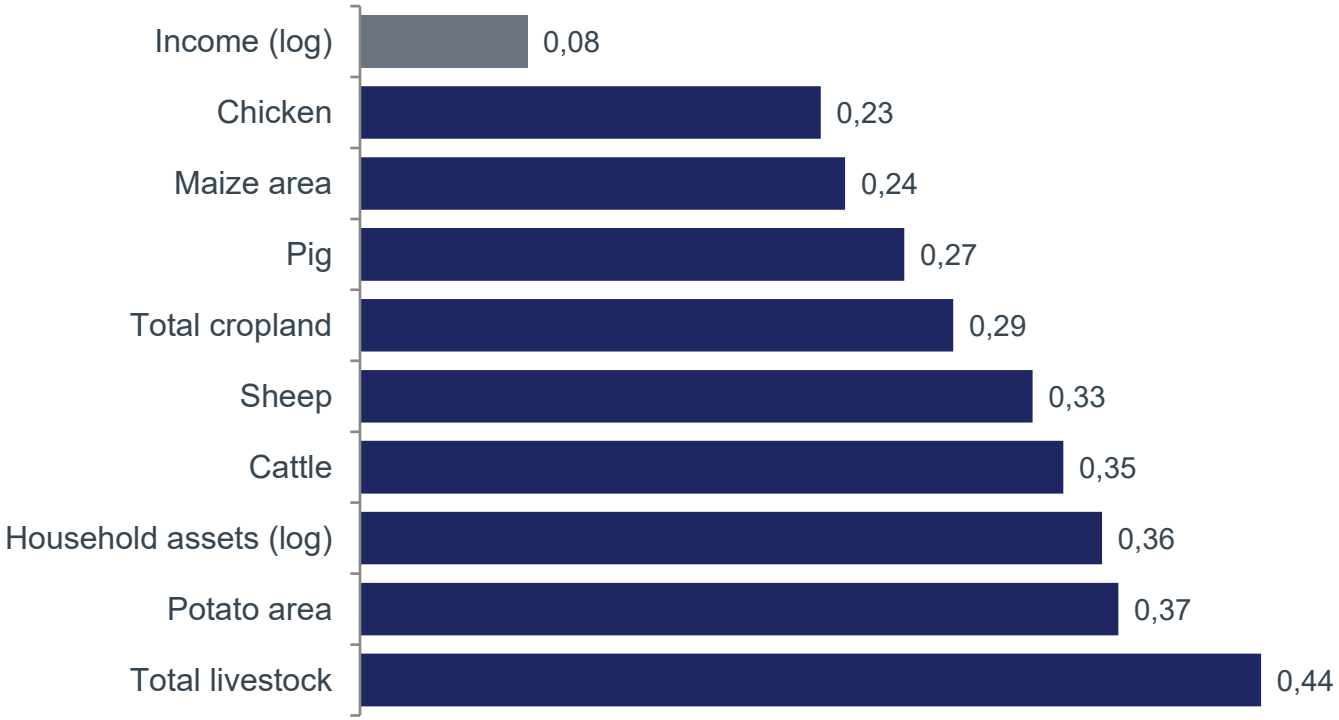
- **Tol1:** support for non-lethal management strategies (e.g. relocation, fencing) - positive framing
- **Tol2:** agreement that government should control wild boar population - reverse encoded, higher raw score = lower tolerance
- **Tol3:** perceived personal safety threat from wild boars - reverse encoded
- **Tol4:** willingness to accept some damage caused by wild boars - positive framing
- All four items use a 1–5 Likert scale where after reverse encoding 1 = low tolerance and 5 = high tolerance
- The four items are not interchangeable; they capture management preferences, policy attitudes, fear, and damage acceptance separately, which is why tolerance is treated as a multidimensional construct in the model

BACKUP

A Household Wealth Index the Original Model Never Used

PCA on 15 crop-area, livestock-count and asset variables; PC1 tracks physical assets, not cash income

PC1 loadings (top variables)



27.9%
of variance explained by PC1 alone (15 input variables)

r = 0.75 with total household assets
Strong — PC1 captures the physical-asset dimension of wealth

r = 0.17 with annual cash income
Weak — income and asset-wealth measure different things here

Why it matters: the original WTM model used a single manifest item (crop asset value) as its only wealth control. This PCA index uses the full livelihood picture — crop portfolio, livestock holdings and total assets together — and shows income is a poor proxy for it (r=0.17). It does not predict Tolerance directly (p=.33), but predicts perceived economic loss strongly (β=0.28, p<.001): wealthier households have more at stake and report it.

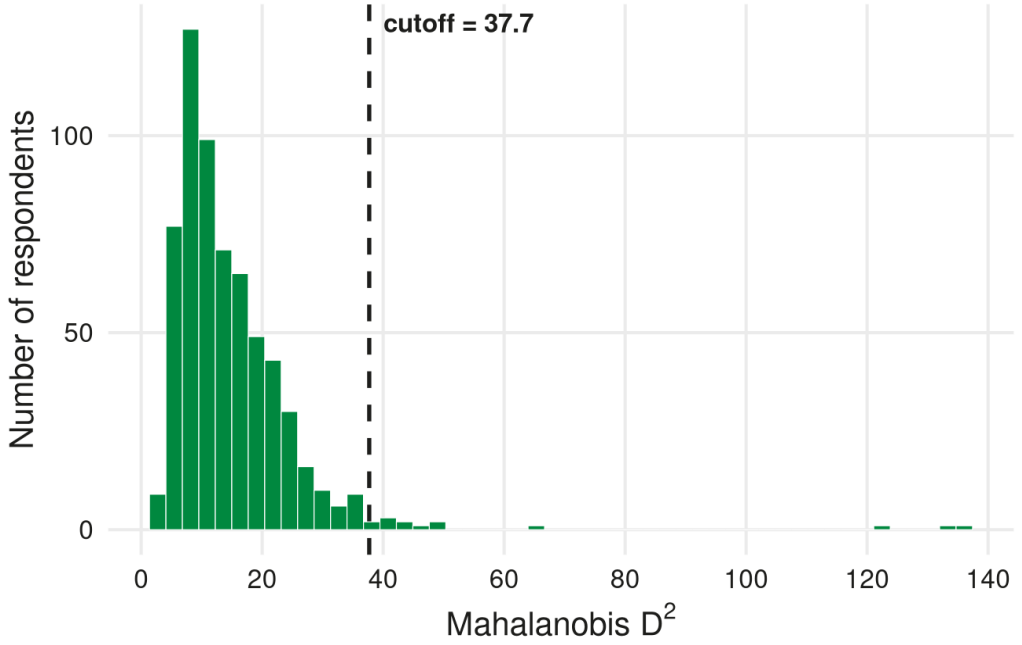
The 14 Multivariate Outliers

Mahalanobis distance across 15 core WTM indicators, with a χ^2-based cutoff

What Mahalanobis distance measures:
for each respondent, how far their combination of scores sits from the "typical" respondent — accounting for how the 15 variables correlate with each other, not just their individual ranges.

The 15 indicators used

Conflict experience (7)	CP_WB1–4, CP_A1–3
Costs (2)	TC1 damaged area, log economic loss
Benefits (2)	IC1 general attitude, IB ecological value
Tolerance (4)	Tol1, Tol2, Tol3, Tol4



Threshold: D^2 values approximately follow a χ^2 distribution with df = 15 (one per variable) under multivariate normality. We flag respondents beyond the p<.001 cutoff: $D^2 > 37.7$.

14 of 625 (2.2%) flagged
Split evenly: 7 Baoxing / 7 Qingchuan, 7 with-compensation / 7 without — not concentrated in either group.

3 respondents are extreme (D^2 = 124–136); the remaining 11 sit just past the cutoff (D^2 = 38–50).

Missing Values Treatment

- Compensation timeliness/satisfaction (column CE1, CE2) are left as NA for the 348/625 (55.7%) who never applied — this is skip-logic, not data loss => these fields are excluded from the full-sample structural model entirely (used only in supplementary with-compensation-only checks).

- Out-of-range compensation percentages (column CA1, CA2) are kept, not deleted or capped by default — 9/625 CA1 values and 4/625 CA2 values exceed 100% (up to 800%)
  => treated as genuine (a respondent can coherently want more than their assessed loss), with a winsorised (capped- at-100%) version run separately as a sensitivity check — results essentially unchanged.

- Zeros that are real answers were kept untouched: prevention-effectiveness (column PE), channel-count and compensation-anticipation (CA) variables all define 0 as a legitimate response ("used nothing" / "no channels" / "expects nothing").

- Everything else in the analysis dataset is complete — 0 missing values across all other 55 variables for all 625 respondents

How We Tested the County Confound

Pearson's chi-squared test of independence, 2×2 contingency table (county × applied for compensation)

	No compensation	With compensation	Total
Baoxing	41 (expected: 89.1)	**119 (expected: 70.9)**	160
Qingchuan	307 (expected: 258.9)	**158 (expected: 206.1)**	465
Total	348	277	**625**

"Expected" = count we'd see in each cell if county and compensation status independent. The test measures how far the observed counts are from that.

$\chi^2 = 77.09$, df = 1

$p < 2.2 \times 10^{-16}$ (i.e. $p < .001$)

Odds ratio = 5.64 Baoxing residents have 5.6× the odds of having applied, versus Qingchuan.

Cramér's V = 0.35 (moderate-to-strong association for a 2×2 table)

Robustness of the test itself:

- Reported figure uses Yates' continuity correction (R default for 2×2 tables); uncorrected $\chi^2=78.72$ — same conclusion.
- Fisher's exact test (no large-sample approximation): $p=5.8 \times 10^{-19}$ — confirms it.

BACKUP

The Five Measured Mediators Behind RQ1

None individually significant, but together they define the indirect path compensation could work through

Code	Construct — survey items	Indirect effect	Bootstrap SE	t-stat	p-value
CA	Compensation Anticipation — CA1 (lowest accepted %), CA2 (expected %), CA3 (insurance WTP, log)	−0.0017	0.0046	−0.36	.800
PS	Policy Support — PS1, PS2	−0.0013	0.0054	−0.24	.808
TC	Tangible Cost — TC1 (damaged area), log(economic loss)	+0.0027	0.0036	0.74	.382
IC	Intangible Cost — IC1, IC2_1, IC2_2, IC2_3	−0.0178	0.0141	−1.26	.165
IB	Intangible Benefit — single item	−0.0060	0.0070	−0.86	.354

Reading the table: each row is one bootstrapped specific-indirect-effect test (Nitzl, Roldán & Cepeda, 2016) for compensation participation → mediator → tolerance, estimated in seminr alongside the direct effect.

Consequence: no single mediator's indirect path is individually significant (all p>.16); summed together the total indirect effect is −0.024, VAF=18% of the compensation→tolerance association — most of the paradox remains unexplained by these five candidates.

BACKUP

Tolerance Item 3 Does Not Belong to the Scale

"Wild boars threaten my personal safety" fails both item-total and CFA tests against Tol1, Tol2, Tol4

Item	Statement	Corrected item-total r	CFA loading (standardised)	CFA z-value	CFA p-value
Tol1	I support non-lethal management strategies	0.618	0.49	(ref.)	—
Tol2	Govt should NOT crack down on boar population (reverse-coded)	0.684	1.04	4.79	<.001
Tol3	**Wild boars threaten my personal safety**	**0.017**	**0.006**	**0.13**	**0.896**
Tol4	I can accept some damage caused by wild boars	0.288	0.35	9.20	<.001

Reading the table: item-total r is Tol-item correlation with the sum of the other three (psych::alpha, n=625). CFA loading is the standardised factor loading from a single-factor confirmatory model on the 4 items (lavaan, WLSMV estimator, ordinal indicators).

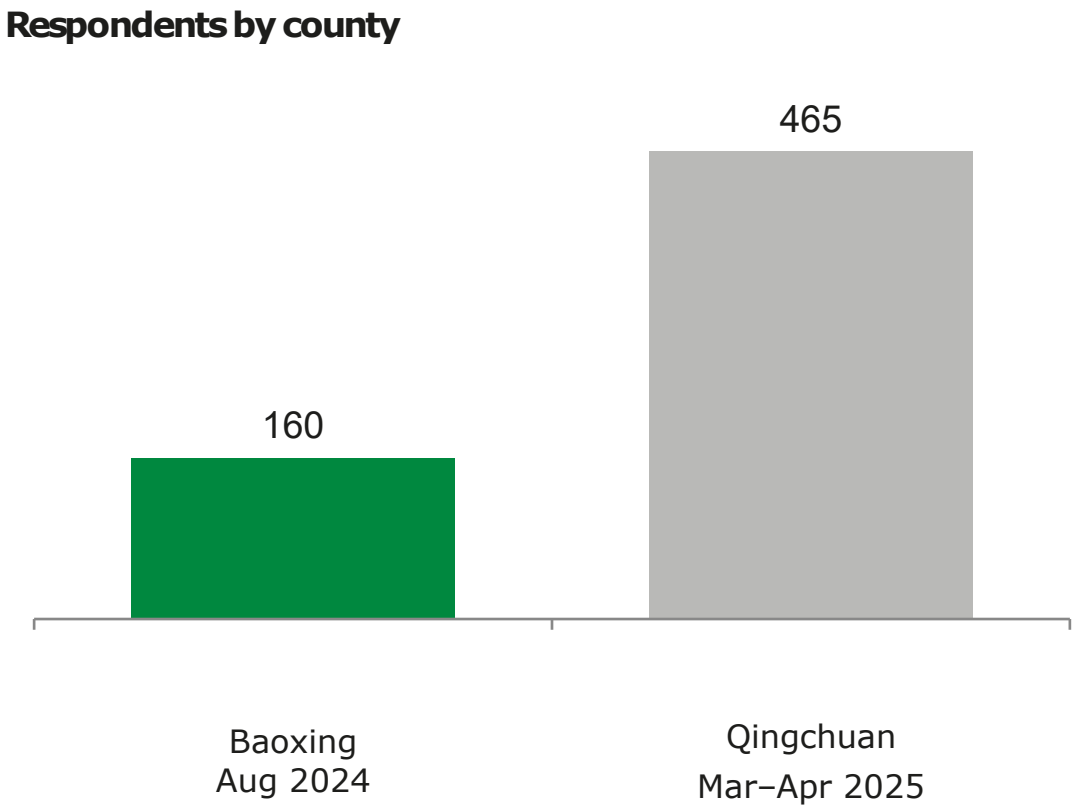
Consequence: dropping Tol3 raises reliability from α = 0.34 to α = 0.50 (ordinal α = 0.62); every downstream structural path (Intangible Cost, Intangible Benefit, Compensation Participation, Prevention Effectiveness → Tolerance) is materially unchanged between the 4-item and 3-item specifications.

How do the distributions look like?

- 625 household respondents across two counties in Sichuan Province, China, both bordering Giant Panda National Park.

- Baoxing county: n=160, surveyed 15–31 August 2024.

- Qingchuan county: n=465, surveyed 10 March – 10 April 2025 — seven months later.



Compensation Experience

Applying and receiving payment are related, distinct questions

- Compensation applicants: 277 self-reported applicants and 348 non-applicants.

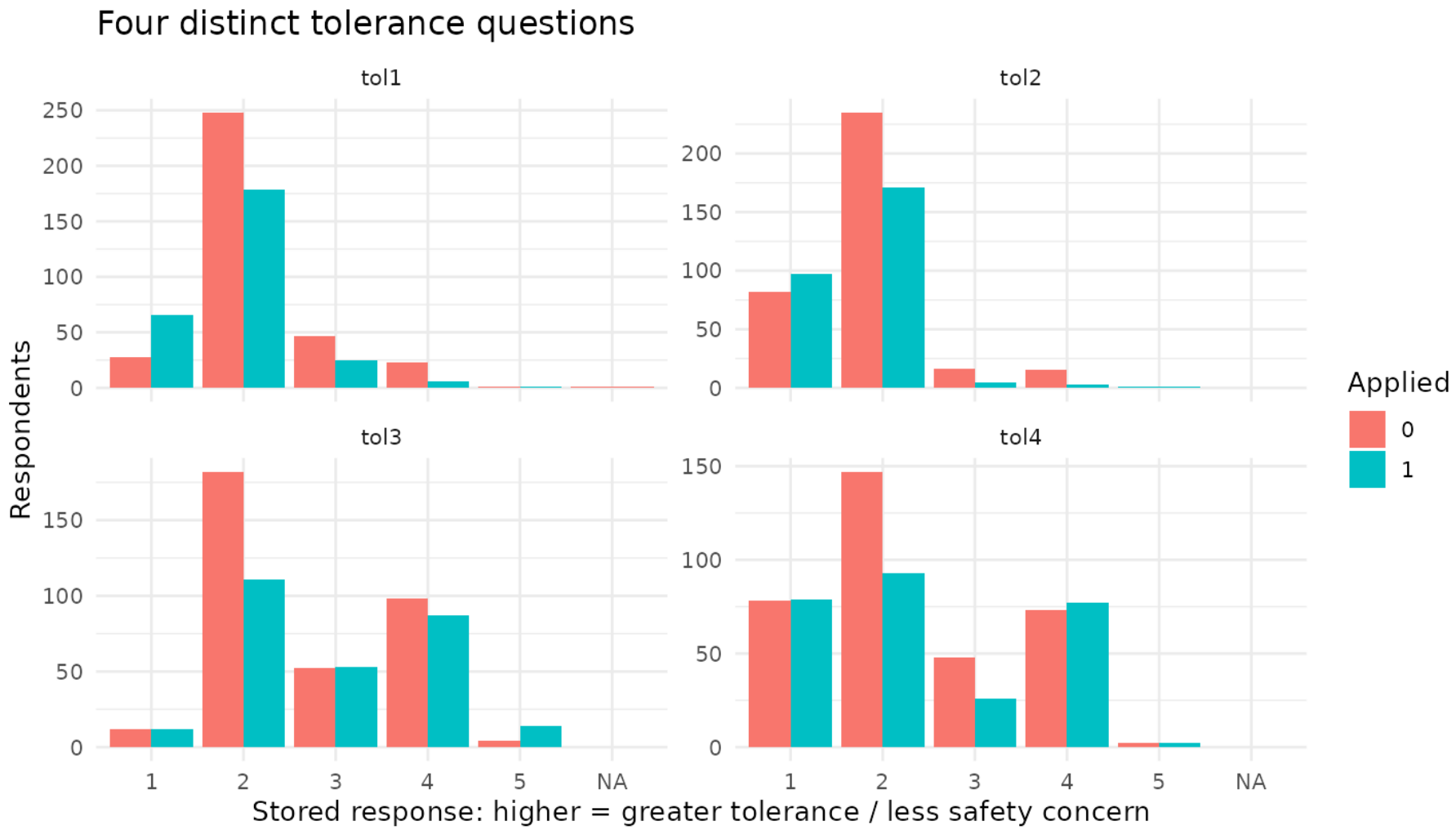
- Among 277 applicants:
  - 275 answered the timeliness question CE1: Was the compensation paid promptly?
  - 276 answered the satisfaction question CE2: Were you satisfied with the payment of compensation?

--> suggest payment experience, but answering them alone does not establish receipt for everyone.

Compensation Experience (CE)

- CE0 identifies 277 self-reported applicants and 348 non-applicants - main grouping variable throughout
- Among the 277 applicants: 275 answered the timeliness question (CE1) & 276 answered the satisfaction question (CE2)
- CE1 and CE2 suggest payment experience but do not confirm receipt for all applicants - answering them alone is not sufficient evidence of actual payment
- CE0 as the primary compensation variable: a clean binary that applies to the full sample

The 4 questions measuring Tolerance



(A small number of) Multivariate Outliers

14

of 625 respondents
(2.2%)

• Mahalanobis distance across 15 core Wildlife Tolerance Model (WTM) indicators (conflict experience, costs, benefits, tolerance items), threshold $p < .001$.
=> Kept in the primary analysis — a statistical outlier is not automatically a data error.

• Every core structural result is re-estimated excluding them as a robustness check (see Robustness section): all paths stable in sign, magnitude and significance.

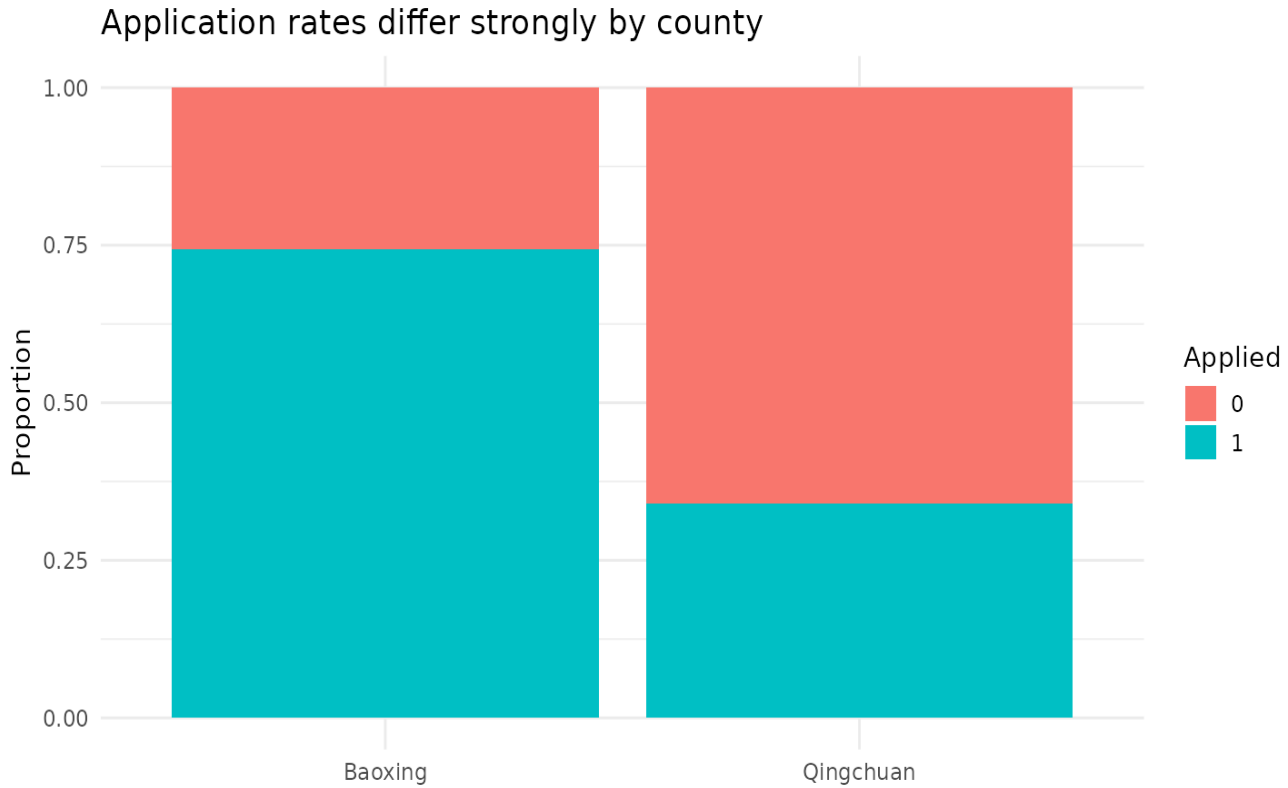
EDA

COMPENSATION YET?

- In Baoxing, 119 of 160 respondents reported applying for compensation.

- In Qingchuan, 158 of 465 reported applying — so geography strongly changes the group composition.

- We therefore compare groups with county included in the adjustment set.



Preparing for modelling: Feature engineering

- **Skewed monetary/area variables are log-transformed** using log(1+x) to keep zero-valued households in the model: household income, household assets, economic loss from damage, insurance willingness-to-pay, and damaged crop area => without this, a handful of very large landholders/losses would dominate every linear relationship involving them.

- **Tolerance is engineered as a 3-item composite (Tol1, Tol2, Tol4), not the original 4**, based on the measurement evidence: item Tol3 ("wild boars threaten my safety") is statistically unrelated to the other three (item-total r=0.02, CFA loading=0.006)
  => The 4-item version is kept alongside as a named robustness variant, not discarded — every core structural result is checked against both and is materially unchanged.

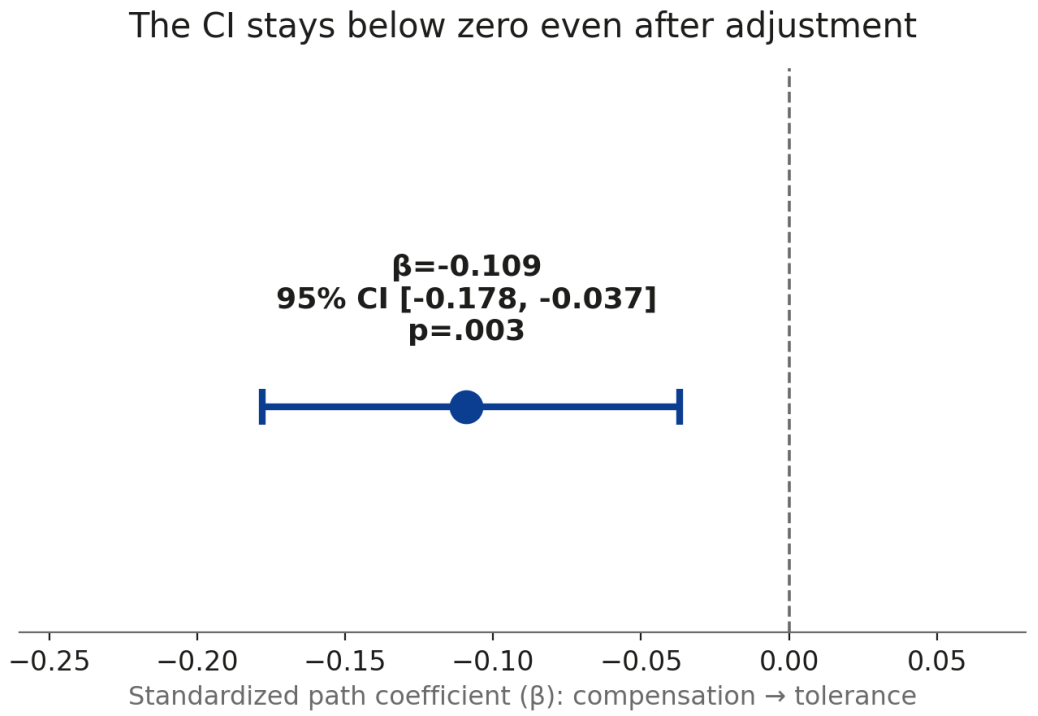
- **New household wealth/livelihood index is engineered from 15 previously-unused variables** (crop areas by type, livestock counts by type, land/livestock/total assets) via **PCA** — the first component explains 27.9% of variance and correlates strongly with total assets (r=0.75) but only weakly with income (r=0.17) => showing income and asset-wealth are not interchangeable here
  => modelling can use a richer wealth measure than the original single manifest item (crop asset value alone).

- **Survey site is engineered as an explicit covariate** (Baoxing vs Qingchuan dummy), because EDA showed compensation participation (column CP) is confounded with county (74% vs 34%) and the two counties were surveyed seven months apart
  => without this feature, **county differences** would be **invisible** to the model and could be misattributed to compensation or conflict experience.

FINDINGS

The compensation paradox remains after site adjustment

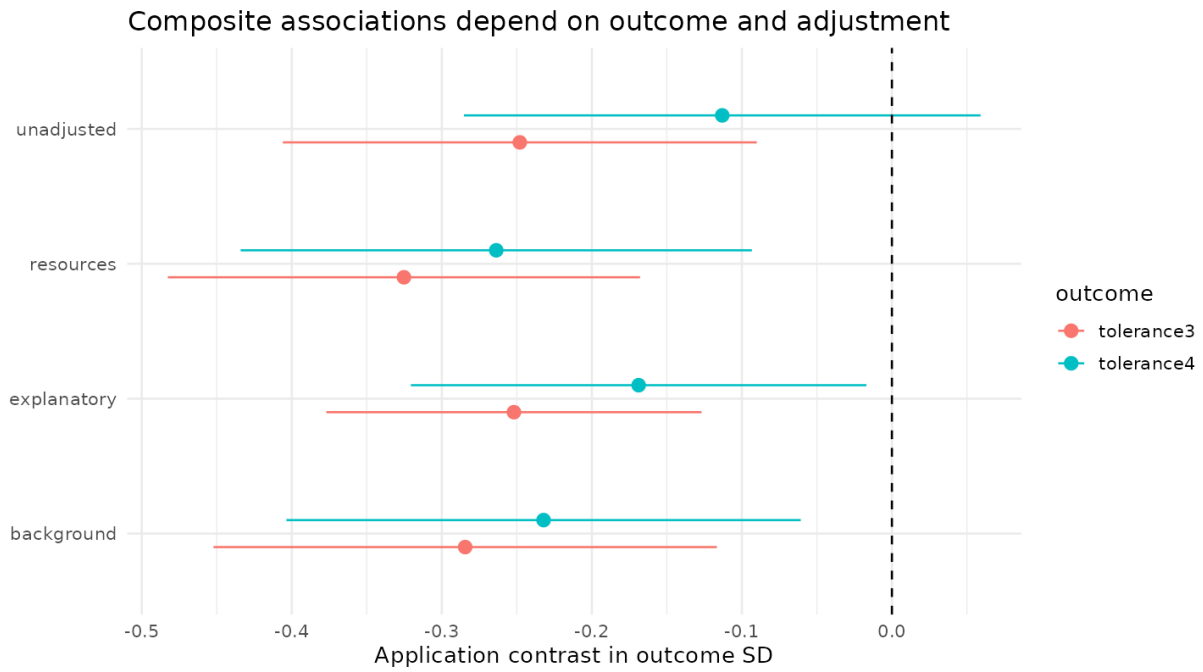
- Compensation participation → tolerance: β = −0.11, p = .003 in the full model.
- Model R^2 for tolerance is 0.234 (adjusted 0.214): meaningful, but much remains unexplained.
- This is an adjusted association, not proof that compensation itself lowers tolerance.



FINDINGS

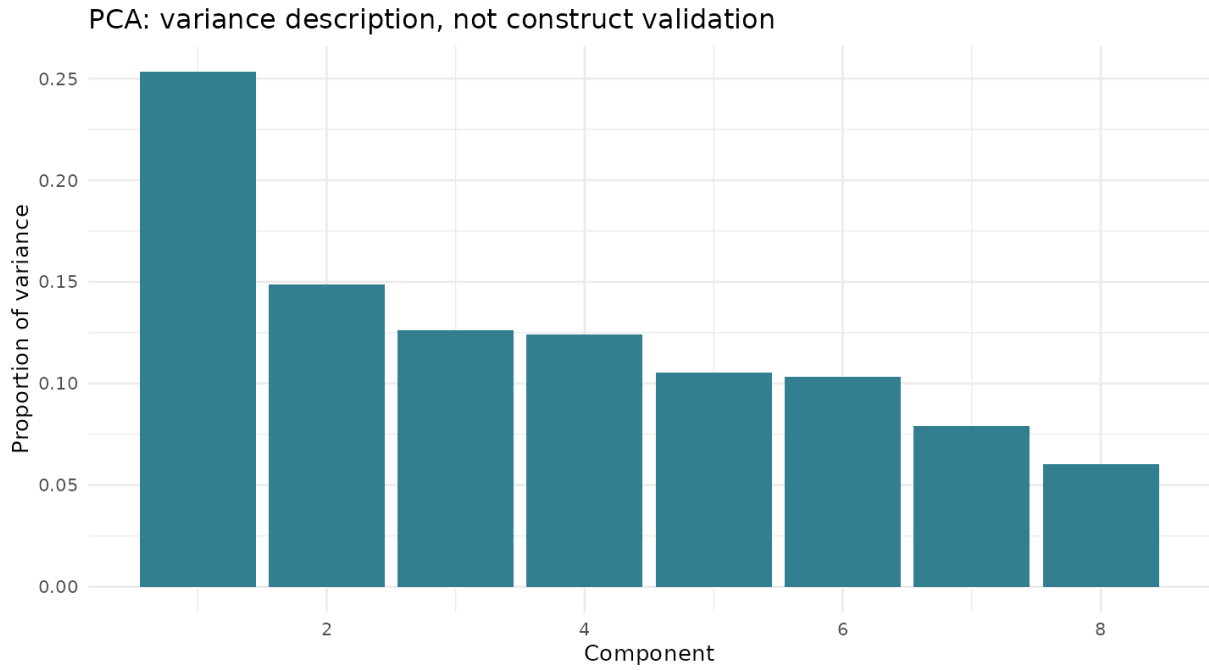
The compensation paradox remains after site adjustment

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- Model R^2 for tolerance is 0.234 (adjusted 0.214): meaningful, but much remains unexplained.
- This is an adjusted association, not proof that compensation itself lowers tolerance.



PCA adds a livelihood index; it does not define a psychological construct

- A PCA of 15 crop, livestock, asset and income variables yields a wealth/livelihood component explaining 27.9% of variance.
- PC1 correlates strongly with total assets (r = .75) but weakly with income (r = .17).
- It predicts reported economic loss, not tolerance; it is a socioeconomic control, not a new latent attitude.



RQ2 finding: intangible costs and benefits matter, but interactions do not

- Intangible cost remains strongly negative; intangible benefit remains positive in the moderation model.

- Tangible-cost × compensation: p = .216; intangible-cost × compensation: p = .234; intangible-benefit × compensation: p = .899.

- The data do not support different cost/benefit slopes for residents with versus without compensation experience.

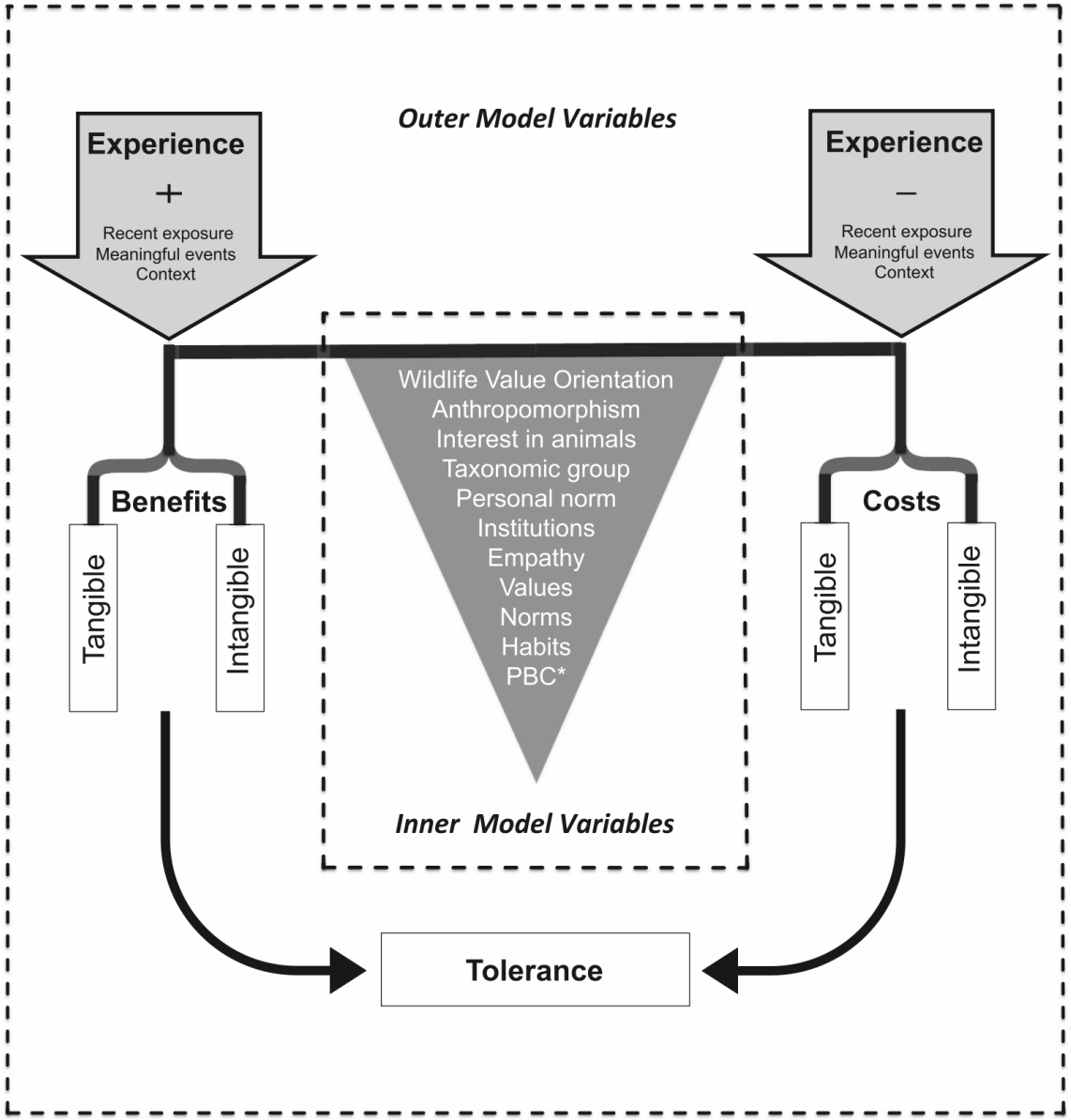
Retaining controls checks whether the cost association survives

- We keep age, gender, education, income, assets and conflict variables in fuller observed-score models.
- The cost association stays negative in both groups, using three or four tolerance items and two cost definitions.
- This supports a persistent association, but does not prove superiority over every measured predictor.

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THE WTM

The original WTM questionnaire design specifies five multi-item constructs as formative by default: Conflict Experience, Tangible Cost, Intangible Cost, Compensation Anticipation, and Policy Support



Kansky, R., Kidd, M., & Knight, A. T. (2016). A wildlife tolerance model and case study for understanding human wildlife conflicts. Biological Conservation, 201, 137-145. https://doi.org/10.1016/j.biocon.2016.07.002

Wildlife Tolerance Model (WTM)

Outer Model

Inner Model

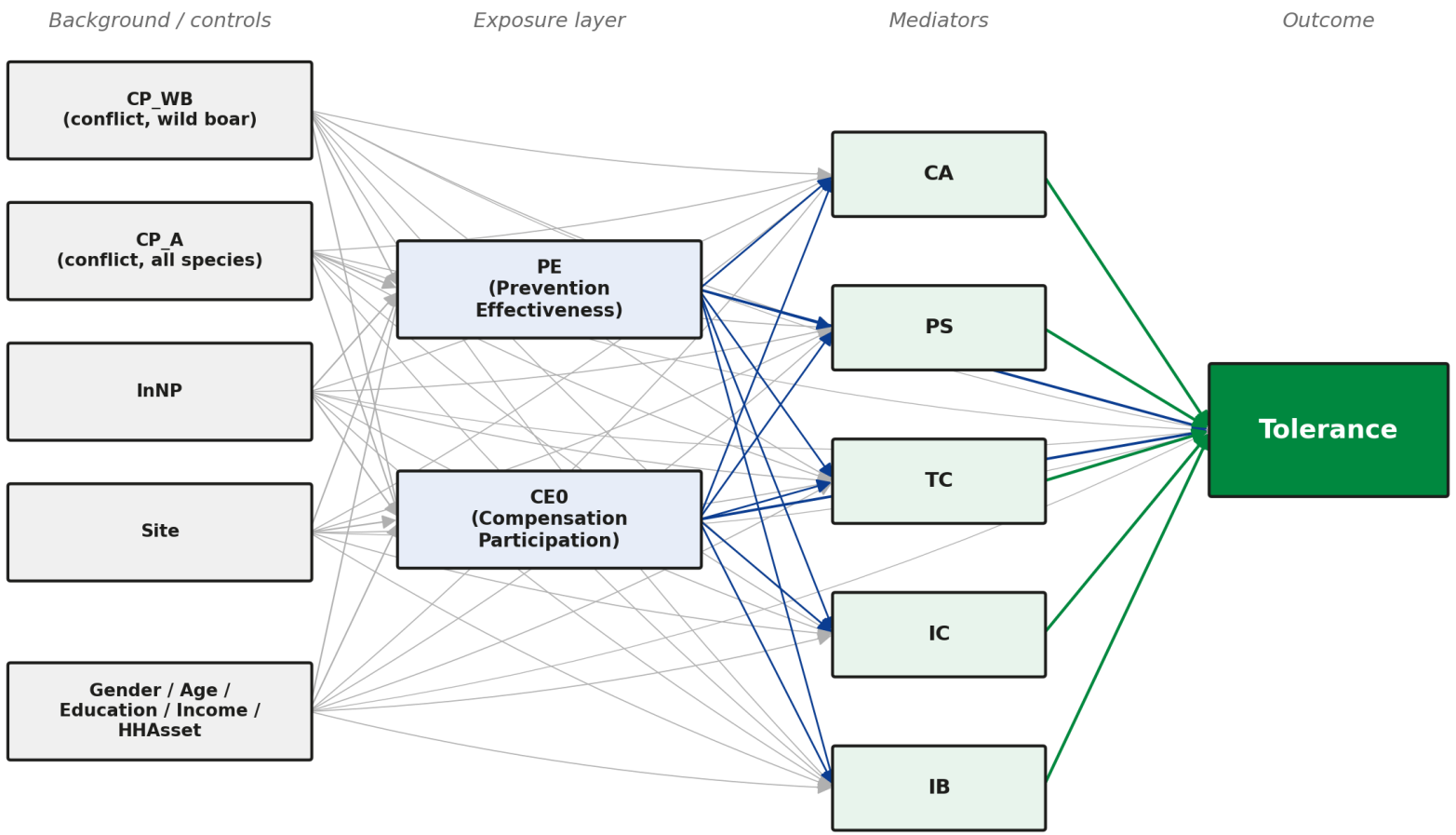
Policy perspective?

Our aim is to see the relationship between compensation policy and tolerance.

BACKUP

Backup for Slide 17: The Full Model Structure

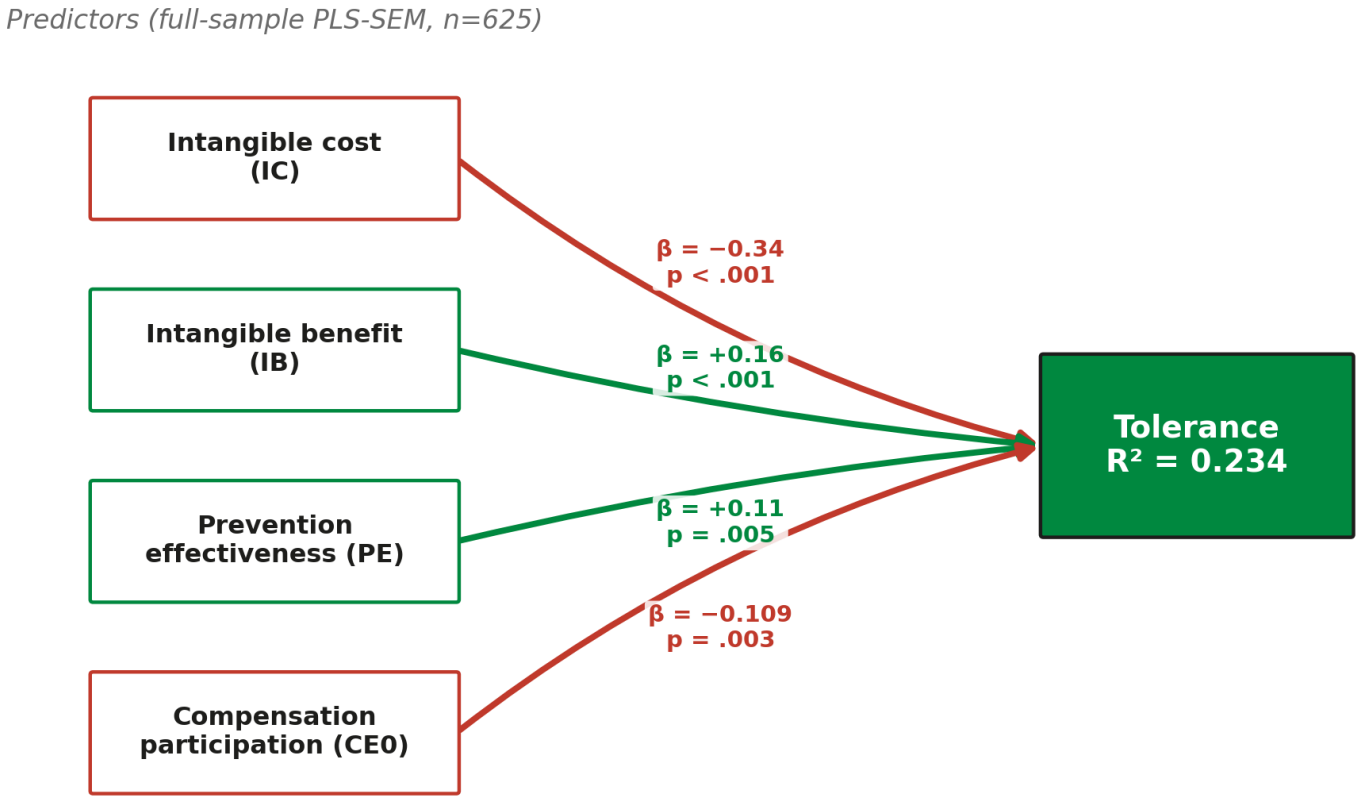
Every construct and structural path in the primary PLS-SEM (n=625)



BACKUP

Backup for Slide 18: RQ1 Direct Effects, Visualized

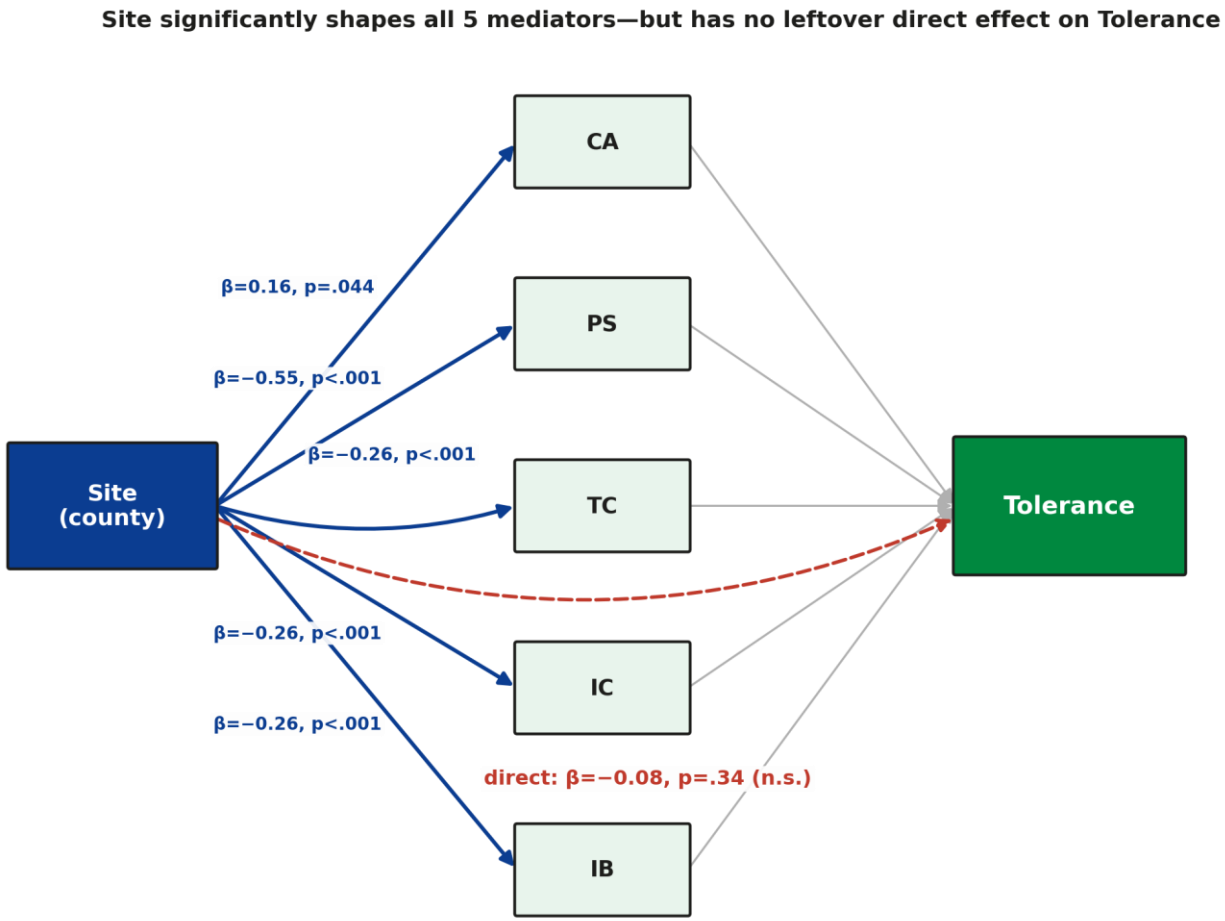
The four direct predictors of tolerance, with standardized path coefficients



BACKUP

Backup for Slide 19: How Site Enters the Model

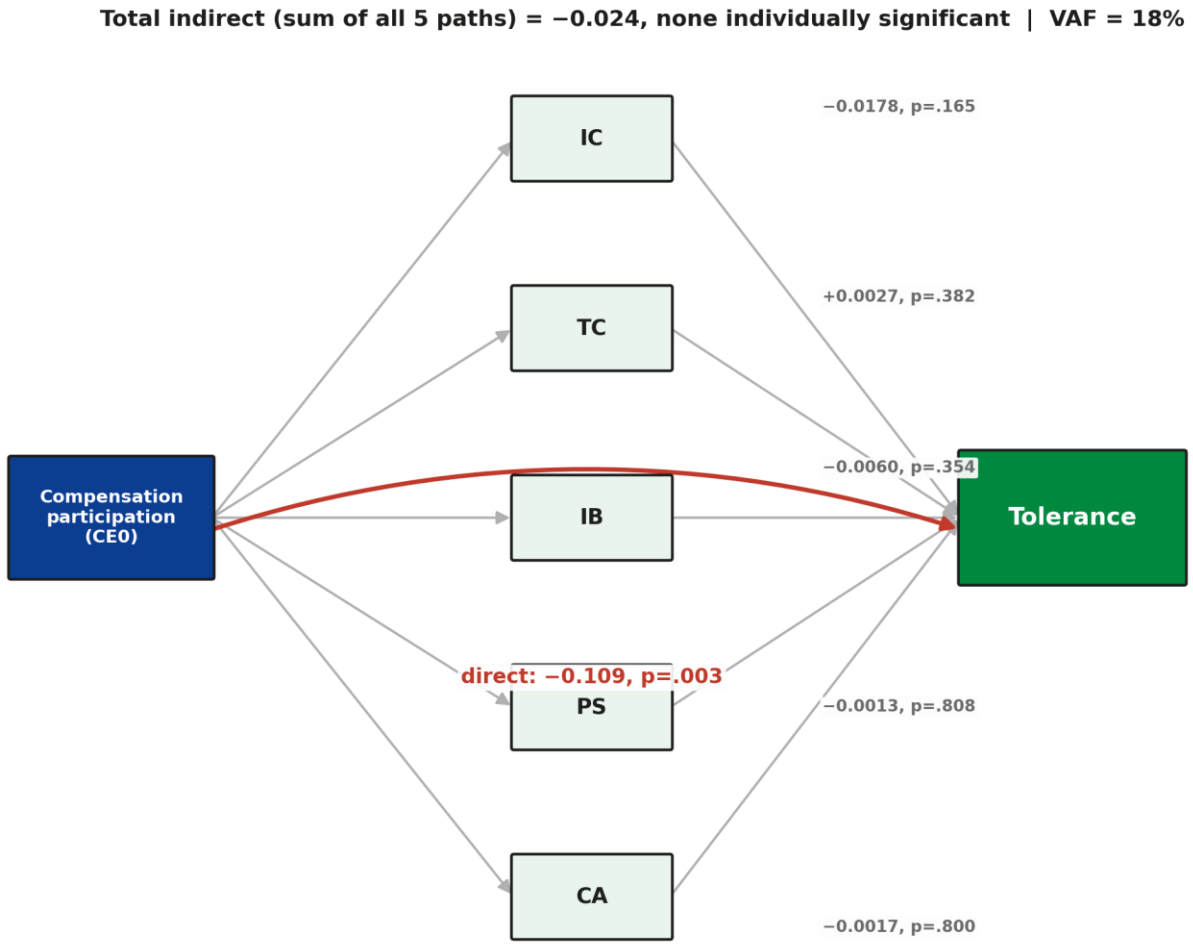
Site's paths to the five mediators, and its non-significant direct path to tolerance



BACKUP

Backup for Slide 20: RQ1 Mediation, Visualized

Direct and indirect (via five mediators) paths from compensation participation to tolerance



BACKUP

Backup for Slide 21: RQ2 Moderation, Visualized

Testing whether compensation participation moderates the cost/benefit → tolerance paths

